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Microstructural and Surface Property Evolution in AISI 2205 and 304L Steels Treated by SMAT and Plasma Nitriding

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ABSTRACT
This study investigates the surface properties of AISI 2205 and AISI 304L stainless steels treated by surface mechanical attrition (SMAT) and plasma nitriding. In SMATed AISI 304L, intersecting mechanical twins formed submicron rhombic blocks within the surface layer; this feature was not observed in SMATed AISI 2205. Surface hardness was significantly enhanced post-SMAT, increasing by about 70–80% in AISI 2205 and over 100% in AISI 304L. However, passive film formation differed: the film on AISI 304L was less stable than that on AISI 2205 under high electric potential and plasma conditions. Variations in passivation behavior between SMATed and non-SMATed steels affected their plasma-nitriding responses

KEYWORDS : SMAT, Corrosion, passivation, attrition, ECAP

Introduction

Nanocrystalline grain-structure at the surface of the alloys is advantageous in enhancing the surface properties. Nano- or submicron-size grains can improve the corrosion resistance of the alloys [1]. Severe plastic deformation (SPD) is one of the processing routes to achieve the nanostructure in the alloys. Various SPD methods like equal channel angular pressing (ECAP), high-pressure torsion (HPT), accumulative roll bonding (ARB), hydrostatic extrusion (HE), surface mechanical attrition treatment (SMAT), ultrasonic shot peening (USP), etc are known for the grain refinement [1,2]. Out of these methods, SMAT and USP are the surface-treatment methods. SMAT is becoming popular in recent years because of its capability to alter the surface properties of various alloys [1,3–9].

SMAT causes the grain-refinement of the metal surface to the nanometre level without affecting its chemistry. Accumulative strain at the surface causes the grain-refinement in this treatment [7]. The refined grains and increased defects affect the passivation tendency of the treated surface [8,9]. Plastic deformation at the microstructural level of a material induces dislocation structures, twins, cell structures, deformation bands and slip systems [10]. A high fraction of the grain-boundaries, dislocation density, and other structural defects increase the stored energy in the material. As per the classical theory of corrosion, this stored energy increases the reactivity of the material with the surrounding environment and accelerates the corrosion by generating numerous micro-electrochemical cells between grain-boundaries and the matrix [11]. Corrosion resistance of the stainless-steels is due to the Cr-rich oxide layer, which acts as a barrier for the diffusion of ions from the surrounding environment. Considerable research work reports about the passivation behaviour of the nano-crystalline stainless steels [8,9,12]. Several studies on the cold-deformed stainless-steels describe the contradictory reports on the corrosion behaviour with an increase in the degree of cold-work [5, 8-13]. Plasma-nitriding is one of the popular methods to enhance the surface properties of technologically important alloys [11,14–16]. Many researchers have recently focused on an approach of duplex surface treatments of stainless-steels, where SMAT or shot-peening was followed by the plasmanitriding [8,9,11,17,18]. The high Cr content helps to improve the surface hardness of the nitriding of stainless steel. The refined nanocrystalline grains by SMAT contain high chemical activity, which can enhance the atomic diffusion of nitrogen in the nitriding treatment to get a

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thicker nitrided layer and increased surface hardness [11]. However, if the oxide layer (passive layer) formed on the stainless steel is not removed, it restricts the growth of the nitride layer by impeding the nitrogen alloying on the surface. Therefore, the growth kinetics of the nitride layer depends on the stability of passive layer and the process parameters that control its removal to facilitate the incorporation of nitrogen in the surface.

The electrochemical behaviour, the characteristics of passive film (like thickness, composition, electronic structures, etc.), and the response of surface to the SMAT and plasma-nitriding can be dependent on the chemistry of stainless-steels. This study focuses on the detailed understanding of the effect of SMAT on the surface behaviour of commercially important stainless steels, AISI 2205 and AISI 304L.

Experimental

In this work, AISI 2205 (duplex) and AISI 304L (austenitic) stainless steels were used. Chemical compositions of the steels are as follows:

- AISI 2205: 0.027 wt.% C, 1.57 wt.% Mn, 22.95 wt.% Cr, 5.60 Ni wt.%, 3.08 wt.% Mo, 0.26 wt.% Si, 0.025 wt.% P, 0.005 wt.% S, and Fe as the balance.
- AISI 304L: 0.024 wt.% C, 1.09 wt.% Mn, 18.55 wt.% Cr, 8 wt.% Ni, 0.41 wt.% Si, 0.023 wt.% P, 0.008 wt.% S, and Fe as the balance.

Sheets of both steels were solution annealed at 1050–1100°C for 2 h. Specimens were quenched in water to avoid the formation of any undesired phases. Specimens of dimensions 70 mm (length) × 20 mm (width) × 4 mm (thickness) were prepared from the solution annealed sheets. SMAT of these specimens was performed using the in-house developed setup [17]. This setup was based on the concept of electromagnetic vibrations. Table 1 shows the specimen conditions used in this work. Specimens were mirrorpolished before SMAT using the standard metallography techniques. Metallographically prepared cross-section of the SMATed specimens was etched using 25 mL HCl + 25 mL H_2O + 4 g $CuSO_4$ solution for 30 s. Such prepared specimens were subjected to the microstructural investigations using optical microscopy and SEM. The microhardness measurements of the specimens were carried out at 100 g load with a dwell time of 10 s. The average of 10 microhardness measurements is reported in this study.

Non-SMATed and SMATed specimens were plasma-nitrided at 500°C for 10 h using 1:4 N_2:H_2 gas ratio. Specimens were sputter cleaned in H_2 atmosphere for 30 min.

Table 1. Designations for the non-SMATed and SMATed AISI 2205 and AISI 304L specimens.

Specimen condition	Specimen designation: AISI 2205 steel	Specimen designation: AISI 304L steel

Open circuit potential (OCP) and potentiodynamic polarisation were used to evaluate the electrochemical behaviour of the SMATed and non-SMATed specimens. The electrochemical corrosion evaluation was carried out in 3.5% NaCl solution using a three-electrode cell, where the stainless-steel specimen was acting as a working electrode, platinum as an auxiliary electrode and the saturated calomel electrode (SCE). The specimens were immersed in the 3.5% NaCl solution for 1800 s to get the open circuit potential (OCP).

The XPS measurements were carried out using photons from an Al target (Kα line, Thermo Alpha 110 Hemispherical Analyser) at 1486.6 eV. The work function was calibrated using a clean in-situ sputtered Ag standard sample and taking in account the BE of the spectral band associated with the $3d^{5/2}$ electrons located at 368.2 ± 0.1 and 1.1 ± 0.1 eV full width at the half maximum (FWHM). The electron inelastic collision background was removed by the Shirley method, and the relative chemical concentrations were obtained using the appropriated sensitive factor given in Refs. [19,20]. Voigt curves were applied to adjust the photoemission spectra bands by considering Gaussian and Lorentzian contributions meeting the following criteria. First, the FWHM Gaussian distribution was fixed in 0.55 eV, value stemming from the XPS analyser resolution used in the CPM (Constant Pass Mode, 20 eV). Second, the Lorentzian width contribution was considered due to both the photon Al target Kα (non-monochromatised) line-width (~ 0.85 eV) and local chemical disorder. It is important to remark that the XPS technique probes up to 10 nm in depth [19].

Compositional depth profiling of the specimens, in particular, nitrogen and carbon profiles, were assessed by glow discharge optical emission spectroscopy (GDOES, Horiba GD Profiler 2). The GD-OES technique allows removing the atoms from the surface by sputtering and, after the emission of the excited species, photosensors for a specific wavelength can detect the presence of atoms [21].

Results and discussion

Microstructure and Hardness

Figure 1 shows the microstructure of the cross-sections of non-SMATed and SMATed AISI 2205 and 304L steels. The non-SMATed AISI 304L specimen, ANS, shows the grain boundaries and twins in some grains

Non-SMAT	DNS	ANS
SMAT (Ball size: 3 mm, Number of balls: 130 balls, SMAT duration: 60 min)	DS1	AS1

(Figure 1(a)). Grains of two phases,austenite and ferrite, appear bright and dark, respectively, in the optical micrographs of AISI 2205 specimens, DNS (Figure 1 (d,e)). The near-surface region of the SMATed AISI 304L specimen, AS1, shows a mesh type features (Figure 1(b,c)), which are the mechanical twins generated during SMAT [7,9]. At higher magnification, SEM micrograph reveals the formation of submicron-size

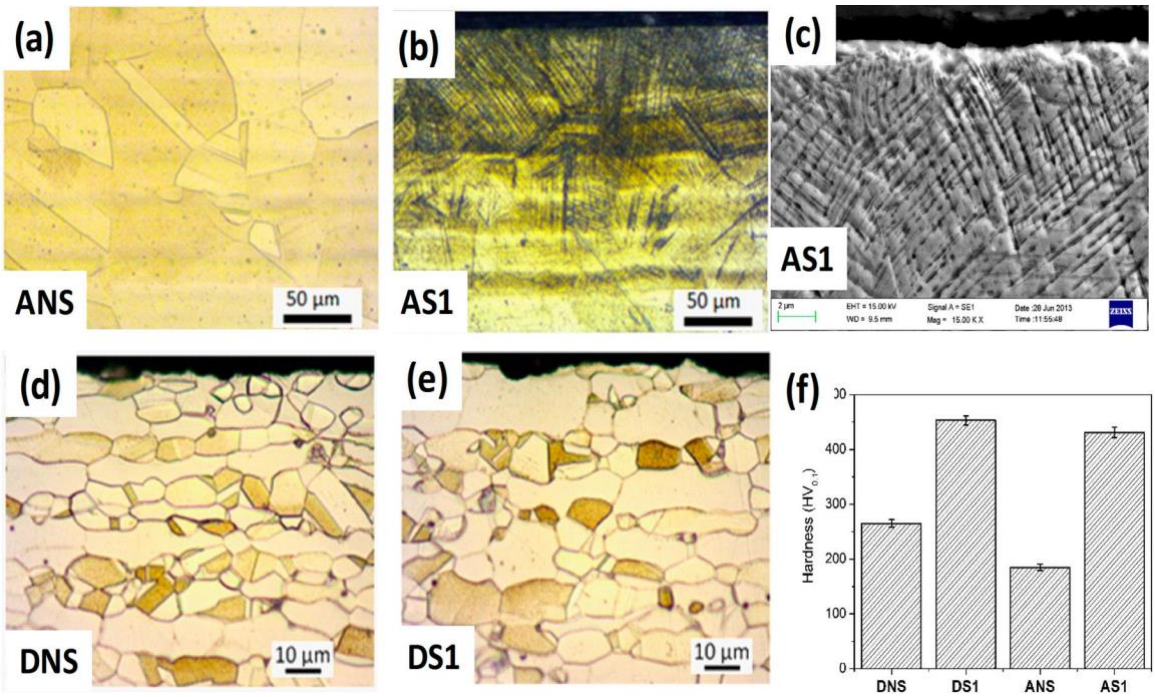

Figure 1. (a)–(e) Microstructures and (f) hardnesses of the non-SMATed and SMATed AISI 2205 (DNS, DS1) and AISI 304L (ANS, AS1)
specimens.

rhombic blocks due to the intersection of the mechanical twins (Figure 1(c)). Such microstructural feature is absent in the SMATed AISI 2205 specimens, DS1 (Figure 1(e)). However, both the steels show considerable enhancement in the surface-hardness due to the SMAT (Figure 1(f)). This improvement in the surfacehardness is about 70–80% for the AISI 2205 steel and more than 100% for the AISI 304L steel. In the SMATed AISI 2205 steel, along with the absence of mechanical twins, the extent of improvement in the surface-hardness is lesser. Generally, highstrain and lowstrain rate promote the formation of twins. The formation of mechanical twins is easier in austenite, γ, phase due to the low stacking fault energy (SFE) [22,23]. Low SFE promotes the formation of more mechanical twins and hence, higher hardness [24]. However, the γphase is relatively difficult to harden in AISI 2205 than in AISI 304L [25]. In AISI 2205, the total deformation distributes among the grains of both phases (γ and α). Typically, SFE of ferrite, α, is higher than the γ-phase. As the deformation increases, the movement of dislocations is easier (than the formation of twins) within the α-phase (due to its higher SFE) [26,27]. Therefore, the two-phase (γ and α) microstructure and SFE differentiate the SMAT behaviour of AISI 2205 steel from the AISI 304L steel. SMAT causes the grain refinement, development of micro-strain and increase in the dislocation density in both steels [8,9,17].

Electrochemical behaviour

Figure 2 presents the results obtained in the electrochemical studies of the non-SMATed and SMATed AISI 2205 (DNS and DS1) and AISI 304L (ANS and AS1) steels in 3.5wt.% NaCl solution. Open circuit potential (OCP) of both steels shifts towards the nobler direction (i.e. less negative value) after SMAT (Figure 2(b)), which suggest the improvement in the corrosion resistance of the steels after SMAT. OCP shift towards the nobler direction indicates the more stable (electrochemically) airborne passive layer (i.e. the oxide layer formed due to the interaction of steel surface with the surrounding air) on the SMATed surface of the steels than the non-SMATed counterparts. This is possibly due to the grain refinement and/or thicker passive layer on the SMATed surface of the steels. However, OCP does not help in understanding the stability of the airborne passive layer (in the solution) in presence of the externally applied potentials. Comparison of the subsequent results obtained from the polarisation study helps in

distinguishing the effect of SMAT and the type of stainless-steels on their surface behaviour.

Figure 2(a) shows the polarisation plots for DNS, DS1, ANS, and AS1. Corrosion potential, E_{corr}, is increased for the SMATed specimens of the steels (Figure 2(c)). The values of E_{corr} is somewhat higher for the non-SMATed and SMATed AISI 2205 specimens than the respective AISI 304L specimens (i.e. DNS versus ANS, and DS1 versus AS1). Passivation current, $I_{passivation}$, is reduced after SMAT of both steels (Figure. 2(d)), which suggests that the SMATed surfaces of both steels have better surface-protection ability (passivation) than the non-SMATed surface in 3.5% NaCl solution. Generally, at higher potentials, the passive film breaks (pitting potential, E_{pitt}), and transition from the passive to transpassive state occurs. Figure 2 (e) shows the variation of E_{pitt} for the non-SMATed and SMATed steels. These values suggest the breakage of the passive film on the SMATed AISI 304L steel occurs at considerably lower potential than its nonSMATed counterpart. This is due to the high surface energy and the defect density introduced by the SMAT process, which produces nanocrystalline grains [11]. However, such behaviour is absent for the AISI 2205 steel, where a marginal change in E_{pitt} is seen after SMAT. These differences in the surface behaviour of the AISI 2205 and AISI 304L steels suggest that the

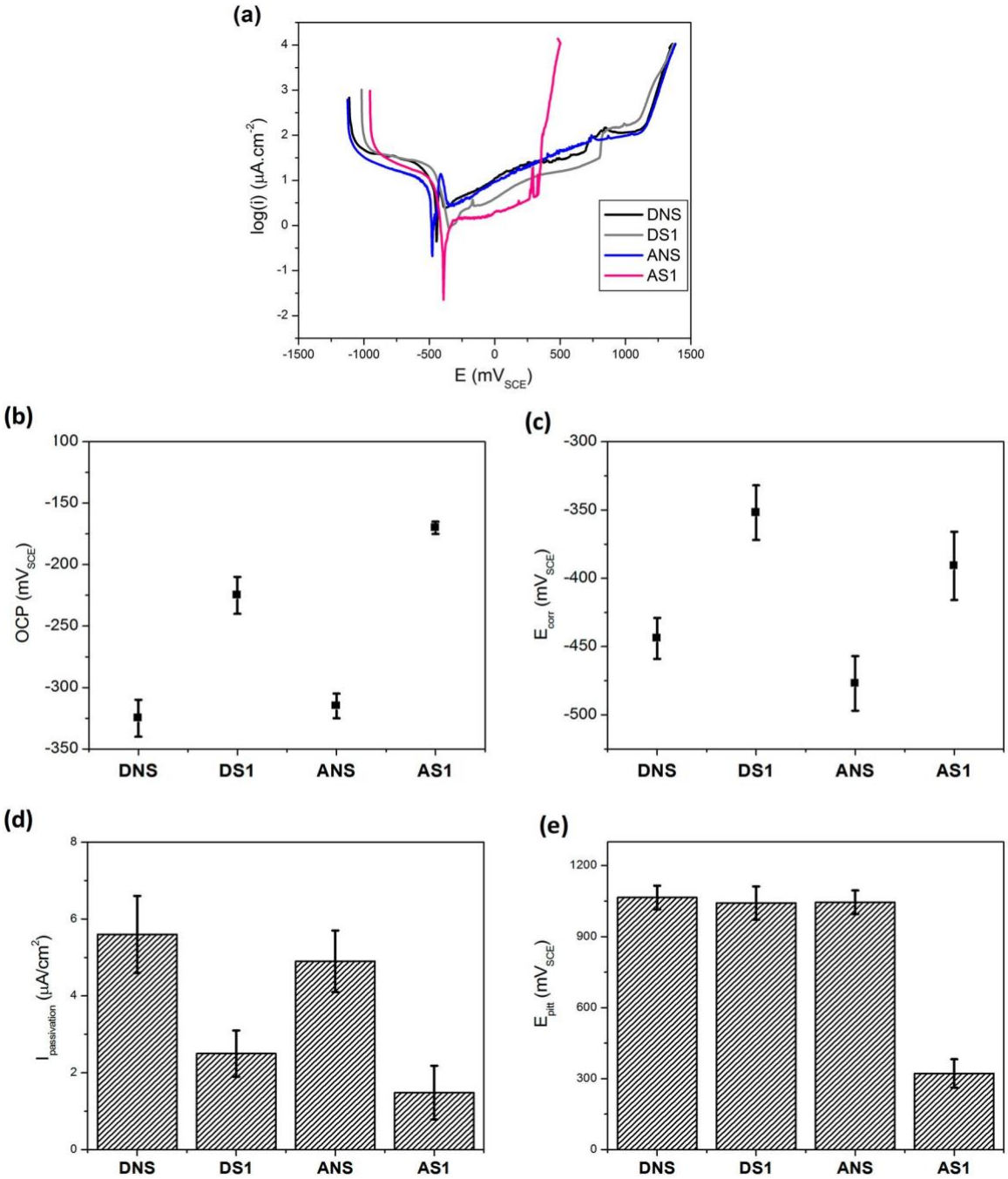

Figure 2. Electrochemical results of the non-SMATed and SMATed AISI 2205 (DNS, DS1) and AISI 304L (ANS, AS1) specimens.

nature of passive film formed on the AISI 2205 steel is different from the AISI 304L steel.

Surface oxidation and its role on plasmanitriding

Figures 3 and 4 show the high-resolution XPS spectra of the non-SMATed and SMATed AISI 2205 and 304L steels, respectively, for the Cr, O, and Fe elements. These results help in understanding the nature of passive film (which is formed due to the interaction of steel surface with the surrounding air). The spectra for Cr and Fe show two peaks. Therefore, to estimate the presence of metallic and oxidation state of these elements, all the spectra are deconvoluted using Voigt curve fitting function within the Shirley background. The deconvolution of Cr(2p) shows the 4 independent peaks located at binding energy of 574.18 (±0.06) and 583.44 (±0.06) eV assigned to Cr($2p_{3/2}$), and 576.11 (±0.20) and 585.73 (±0.35) eV of Cr($2p_{1/2}$) core levels, assigned to the metallic Cr and Cr^{3+} cations, respectively (Figure 3(a,d) and Figure 4(a,d)). The observed energy separation between core levels of 9.21 (±0.05) and 9.62 (± 0.15) eV assigned to the metallic Cr and Cr^{3+} cations [28]. Likewise, deconvolution of Fe(2p) showed 4 peaks located at binding energy of 707.09 (±0.03) and 719.95 (± 0.03) assigned to Fe($2p_{3/2}$), and 709.29 (± 0.05) and 722.15 (± 0.04) assigned to Fe($2p_{1/2}$) core levels of metallic Fe and Fe^{2+} cations, respectively (Figure 3(b,e) and Figure 4(b,e)). Both the steels show the intense peaks for metallic Fe

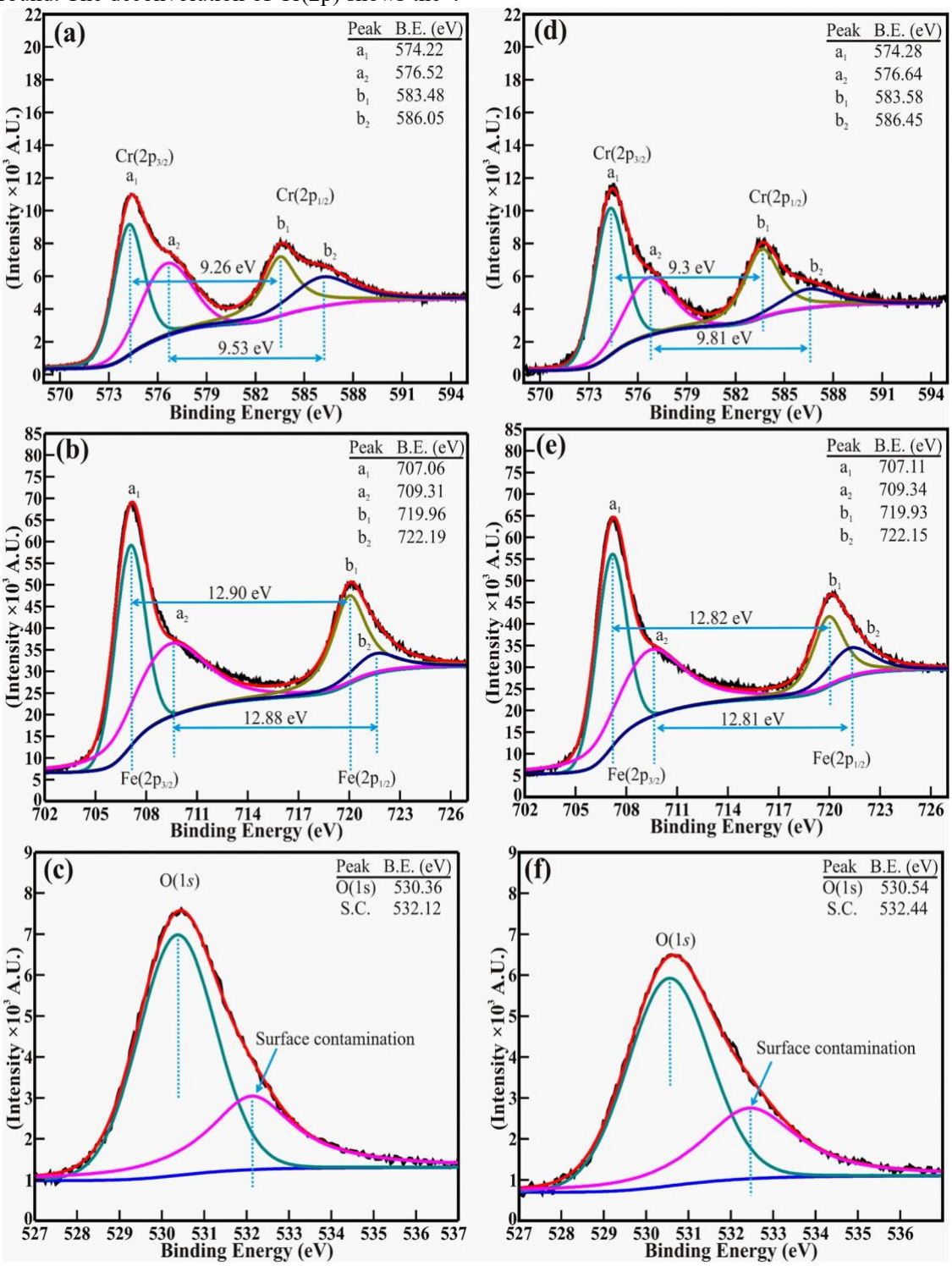

Figure ted and SMATe

Figure 4. Typical high-resolution XPS spectra of the (a, d) Cr (2p), (b, e) Fe(2p) and (c, f) O (1s) core levels of non-SMATed and SMATed 304L steels, respectively. The XPS spectra were decomposed via Voigt curve function fitting.

(Fe– Fe), and very less intense peaks for Fe^{2+} ions, which indicate the presence of Fe, only [28]. However, these steels show the peaks for metallic Cr and oxides of Cr. Typically, the depth profiling for XPS is ~10 nm. Presence of metallic Cr (Cr–Cr) peaks along with the oxides of Cr (Cr–O) suggests that the thickness of the oxide layer is very thin (less than ~10 nm). Considering the affinity of Cr towards the oxygen to form the oxides, the chemical situation of the oxygen is also accessed. Figure 3(c,f) shows the spectra for the O(1s) photoelectron region for the AISI 2205 steel. A predominant peak observed at a binding energy of 530.36 (±0.11) eV is assigned to O(1s) core level. The smaller peak observed at a relatively higher binding energy of 532.49 (±0.22) eV is assigned to the surface contamination due to the OH/C ions [29,30]. As per the literature [31], the former band corresponds to the oxygen associated in the lattice. The surface of the DS1 specimen absorbs more oxygen than DNS (Figure 3(c,f)). The ratio of the peak intensities of the oxides of Cr to the metallic Cr is higher for the DS1 than DNS (Figure 3(a,d)). These results confirm the more Croxide (i.e. thicker oxide layer) at the surface of DS1 than DNS, which gives the better corrosion resistance of the DS1 than DNS (Figure 2). The increased reactivity of the SMATed surface could cause this surface behaviour of DS1.

In case of AISI 304L steel, the ratio of the peak intensities of the oxides of Cr to the metallic Cr is lower for AS1 than ANS (Figure 4(a,d)) which implies that the Cr bounded with oxygen is lower for AS1 than ANS. However, the surface of AS1 absorbs an almost same quantity of oxygen as ANS (Figure 4(c,f)), which indicates that the stoichiometry of Cr-oxide formed on the surface of AS1 is somewhat different than ANS (i.e. if $CrO_{1-\delta}$ forms on ANS, CrO forms on AS1). As mentioned in the previous section, breakage of the passive film on AS1 occurs at considerably lower potential as compared to ANS which indicates the lower stability of Cr-oxide on AS1 than ANS.

The Cr bounded with oxygen and oxygen uptake is higher for ANS than DNS (Figure 4 (a,c) versus Figure 3(a,c), respectively) which implies the thicker oxide layer on ANS than DNS. This phenomenon could be linked with the lower diffusivity of oxygen ions in DNS than ANS. In stainless steels, Cr has an affinity for oxygen to form oxides. Generally, a higher concentration of the reacting element reduces the kinetics of diffusing element [32]. Therefore, the other possibility for the thicker oxide layer on ANS could be due to the lower Cr content in ANS than DNS.

To find out the relative proportion of (free or combined) Fe, Cr and O elements in the passive layer, their respective peaks (i.e. Fe(2p), Cr(2p) and O(1s)) were considered after systematic deconvolution. Figure 5 shows the relative proportion of the considered elements and oxides. The higher relative percentage of oxygen in the DS1 specimen (~22.5%) compared to that of DNS specimen (~15%) confirms the enrichment of oxygen in the SMATed surface of AISI 2205 steel (Figure 5(a)). Since XPS gives the surface information, this higher oxygen content is possibly due to the thicker passive layer on the SMATed surface than its non-SMATed counterpart. This is evidenced in raising the proportion of Cr–O for DS1 than DNS (Figure 5(b)). On the contrary to the SMATed AISI 2205 steel, the SMATed AISI 304L steel shows the marginal change in the enrichment of oxygen and decrease in the Cr–O proportion as compared to the non-SMATed AISI 304L steel (Figure 4(a)). This indicates that the less amount of oxygen is combined with the Cr to form the Cr-oxide on ANS than AS1 (i.e. if $CrO_{1-\delta}$ forms on ANS, CrO forms on AS1).

The differences in the surface behaviour of both steels are possibly linked with their chemical composition, single/two-phase microstructure and the phenomena of mechanical twinning. Overall Cr content in the AISI 2205 steel is more than the AISI 304L steel. AISI 2205 has a two-phase microstructure, where α-phase contains more Cr relative to its neighbouring γ-phase. The enhanced passivity of AISI 2205 steel is due to the greater percentage of Cr in it. Chromium forms insoluble Cr_2O_3 and a continuous network of Cr–O–Cr–O, which helps in preventing the dissolution of iron [33]. The presence and absence of Mo in AISI 2205 and AISI 304L steels, respectively, plays also an important role in governing the passivation behaviour of the steels [33]. Mo helps to form a stable protective layer of Cr-oxides. As mentioned above, the passive layer on DNS has a lower thickness than ANS. Therefore, the Cr-content in the surface oxide per unit thickness of the passive layer could be higher for DNS. As per the literature, presence of Mo increases the Cr content in oxide layer [33]. Owing to the presence of Mo, ionic selectivity in the outer part of the passive film changes from anionic selectivity to cationic selectivity. This promotes the enrichment of Cr_2O_3 in the passive film. Also, it participates in the deprotonation of hydroxide by becoming an electron acceptor, generating oxygen abundance in the inner part of the passive film, and promotes the stability of Cr_2O_3 [33,34].

Both the steels respond differently to the SMAT, and therefore, the microstructure of the SMATed layer differs from each other. Mechanical twins are absent in AISI 2205 steel. However, the near-surface region of the SMATed AISI 304L steel shows a high-density of mechanical twins. The AISI 2205 steel does not show the formation of deformationinduced martensite (α′) or

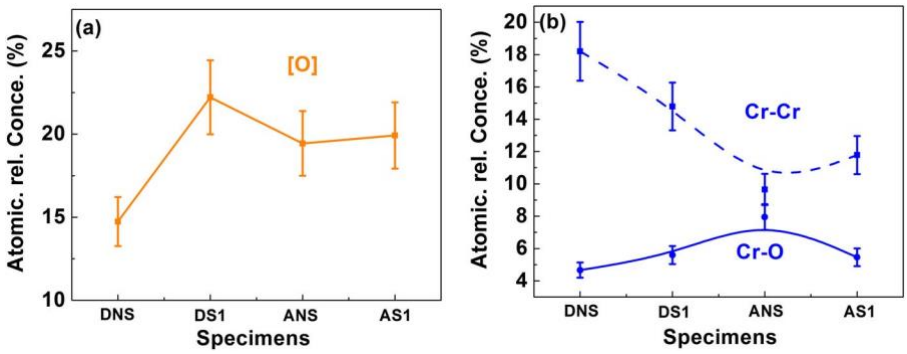

Figure 5. Relative atomic concentration of (a) O and (b) Cr and Cr–O for the non-SMATed and SMATed AISI 2205 (DNS and DS1) and
AISI 304L (ANS and AS1) specimens. The spectra deconvolution uses a symmetrical peak type PsdVoigt 1 (Gaussian and Lorentz

any significant microstructural changes despite the accumulation of strain due to the SMAT (Figure 1(e)). This suggests that the accumulated strain is not sufficient to form twins and to induce the phase-transformation in the AISI 2205 steel. However, the same SMAT-parameters cause the pronounced effect on the microstructure of the AISI 304L steel (Figure 1(a–c)). Deformationinduced martensite, α′, is observed along the twins and at the intersection of the twins in the SMATed surface of AISI 304L steel [9,17]. It is known that the strain affects the formation deformation-induced martensite in the austenitic steels. With increase in strain, the volume fraction of α′ increases [17]. The presence of α′ in the austenite matrix leads to a decrease in the corrosion resistance of the AISI 304L steel [35]. Electron work function of the passive layer generally decreases with increasing plastic deformation. This means that the surface of the metal reacts easily, and increases the metal dissolution. It is reported that increasing strain slightly reduces the corrosion resistance of the duplex stainless steels [36]. However, all such studies have reported the formation of α′ in the deformed duplex stainless steel [13,36]. However, in the current work, the SMATed AISI 2205 steel does not show the presence of α′, and hence, it shows the good corrosion resistance.

All these differences mentioned above are responsible for the different electrochemical behaviour of the SMATed and non-SMATed AISI 2205 and AISI 304L steels.

The role of different passivation behaviours of the SMATed and non-SMATed AISI 2205 and AISI 304L steels is visible in their plasma-nitriding response. Typically, the plasma-nitriding involves two steps: (i) sputter cleaning of the surface using hydrogen or argon, and (ii) supplying the nitrogen along with hydrogen or argon (so that the nitrogen diffuses in
contribution).

the steel surface). Here, the first step should be conducted for a sufficient duration to remove the passive layer (i.e. surface oxides) and contamination. Insufficient duration of the sputter cleaning affects the nitriding kinetics (second step) and reduces the 'effective nitriding time (t_{eff})' [9]. The effective nitriding time is the difference between the total nitriding time (t) and the time required to remove the obstacles for the nitrogen incorporation in the surface ($t_{obstacle}$) [9]. If the duration of the first step is not sufficient, it consumes the part of total nitriding time, and therefore, the effective nitriding time reduces. Duration of the sputter cleaning for the specimens used in the current work is not sufficient, and therefore, t_{eff} is lower than the

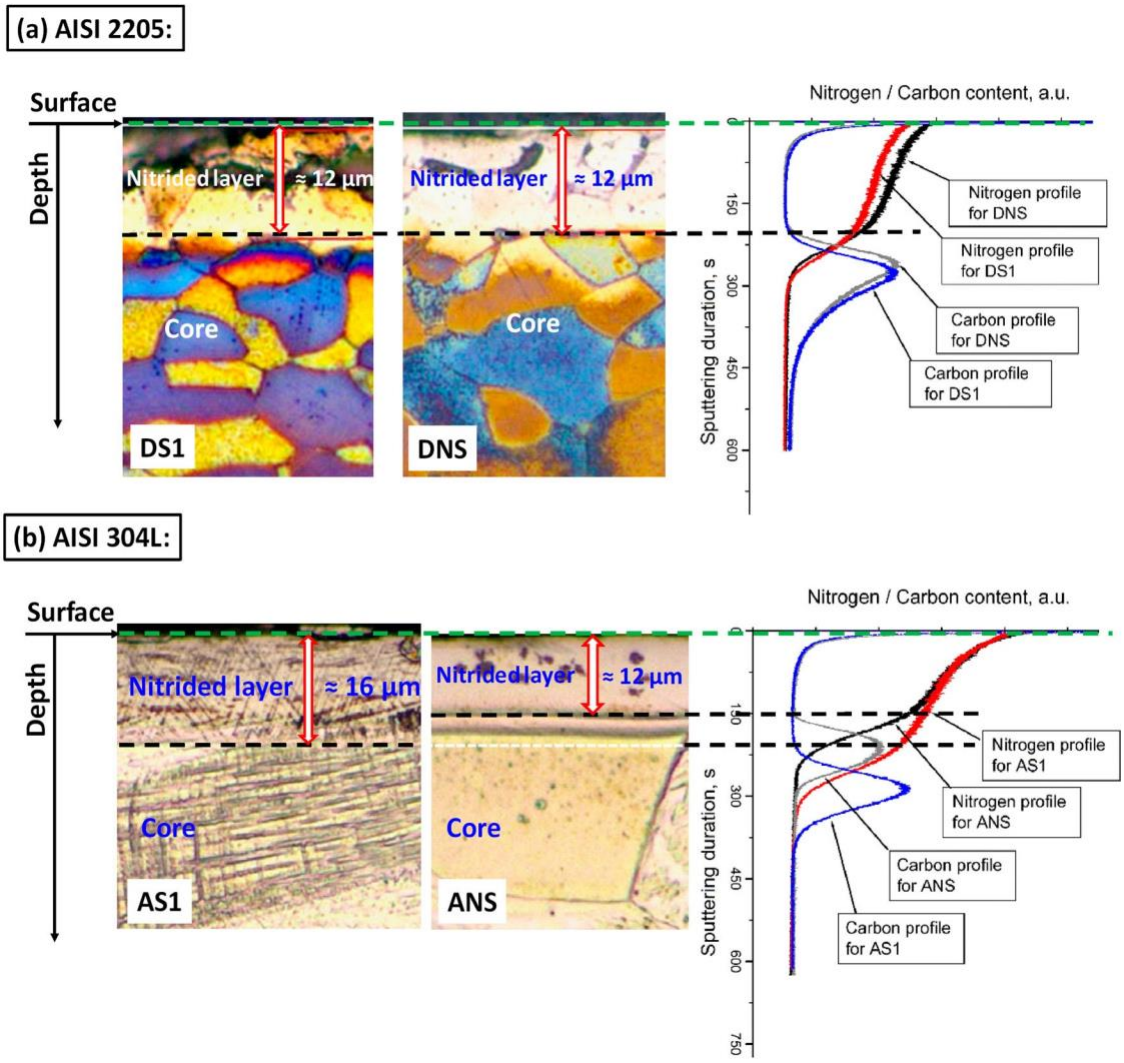

Figure 6. Nitrogen and carbon depth profiles obtained by GDOES and corresponding optical micrographs of the plasma-nitrided non-SMATed and SMATed (a) AISI 2205 steel, and (b) AISI 304L steel.

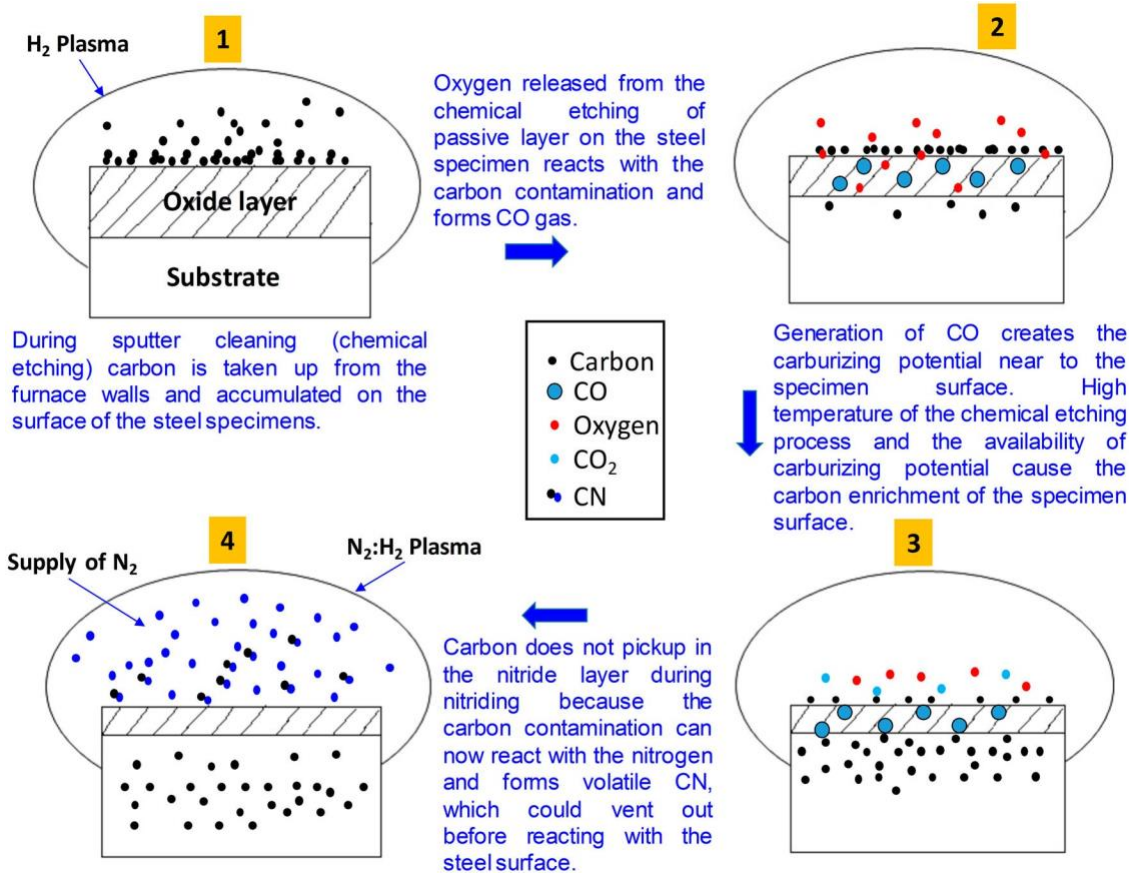

Figure 7. Schematic depiction of the possible mechanism of carbon uptake during plasma-nitriding of stainless steels.

total nitriding time. The extent of $t_{obstacle}$ depends on the nature of passive layer on the specimens.

Figure 6 shows the qualitative results (obtained by GDOES) of the nitrogen and carbon profiles across the depth of plasma-nitrided specimens. Nitrogen uptake in the nitrided layer (near-surface region) is higher than the core of the specimens. As per the literature [7,9], SMAT helps in enhancing the kinetics of surface alloying due to the enhanced crystal-defects in the SMATed layer. However, negligible change in the thickness of the nitrided layer is observed for the AISI 2205 steel (Figure 6(a)). This is due to the more $t_{obstacle}$ for DS1 than DNS. Higher $t_{obstacle}$ for DS1 nullifies the advantage of SMAT in enhancing the nitriding kinetics. Electrochemical study and chemical characterisation by XPS mentioned above confirm the better passivation behaviour (thicker Cr-oxide) of DS1 than DNS (Figure 3), and hence, the passive layer on DS1 requires more time to sputter away during the plasma-nitriding. For AISI 2205 steel, an outermost thicker Cr-oxide layer imparts higher corrosion resistance in detriment of greater nitrogen content in the nitrided layer. However, the nitriding response of AISI 304L steel is entirely different from the AISI 2205 steel. Passive layer on AS1 has sputtered away more easily than ANS (because it contains relatively less stable Cr-oxide in plasma environment). In another word, t_{eff} is more for the AS1 than ANS. Therefore, the nitrided layer is thicker for AS1 than ANS (Figure 6(b)). Moreover, the nitrided layer of ANS is constituted by two different expanded austenite layers as published elsewhere [37].

Presence of carbon is observed at the interface of the nitrided layer and the core of the specimens (Figure 6) even though it is not added intentionally in the plasma-nitriding process. This carbon comes from the furnace wall due to the sputtering action of the ions inside the furnace [9,38]. The possible mechanism for such carbon uptake is described schematically in Figure 7.

Response of AISI 2205 and AISI 304L to SMAT and plasma-nitriding – mechanism

The response of AISI 2205 and AISI 304L steels to the surface treatment is quite different from each other. Unlike AISI 304L steel, SMAT does not cause the formation of mesh-type network of mechanical twins near the surface of AISI 2205 steel (Figure 1). However, SMAT causes an increase in the strain and hardness at the surface. The accumulated strain due to the SMAT is not sufficient to form the twins and to induce the phase-transformation in the AISI 2205 steel.

Grain refinement and/or the defects generated at the SMATed surface help in the formation of more stable (electrochemically) airborne passive layer (which is formed due to the interaction with the surrounding air)

for both steels (as compared to their non-SMATed counterparts) (Figure 2(b)). However, this trend in the stability of passive layer is not followed in the presence of externally applied electric-potentials (Figure 2(e)) and plasma-environment (Figure 6).

SMAT causes the formation of thicker passive layer on AISI 2205 steel than its non-SMATed surface (Figures 3 and 5). Like the non-SMATed surface, this passive layer on the SMATed surface maintains its stability even at elevated electric-potentials (E_{pitt}) in the salt solution (Figure 2(e)). The high Cr content, presence of Mo, and the absence of mechanical twins and α′ play an important role in enhancing the stability of passive layer on the SMATed and non-SMATed AISI 2205 steel (as compared to the AISI 304L steel). In the plasma-nitriding process, sputter cleaning takes more time to get rid of the passive layer on the SMATed AISI 2205 steel (due to the thicker passive layer on the SMATed specimen than the non-SMATed counterpart) and hence, neutralises the advantage of SMAT in enhancing the nitriding kinetics (Figure 6
(a)).

In case of the AISI 304L steel, stoichiometry of the Cr-oxide layer is different for the SMATed and nonSMATed surfaces. The passive layer on the SMATed surface is relatively less stable due to the presence of mechanical twins, formation of α′, and lower Cr content in the oxide. Therefore, it sputters away more easily and enhances the nitriding kinetics in the SMATed surface of AISI 304L steel (Figure 6(b)).

Conclusions

(1) The intersection of the mechanical twins formed the submicron-size rhombic blocks in the SMATed region of AISI 304L steel. However, such feature was absent in case of the SMATed AISI 2205 steel.

(2) SMAT caused considerable enhancement in the surface-hardness of both steels. This improvement was about 70–80% for AISI 2205 steel, and more than 100% for the AISI 304L steel.

(3) SMAT enhanced the corrosion resistance of both steels. However, the overall electrochemical behaviour of their surfaces was not identical. Breakage of the passive film on the SMATed AISI 304L steel occurred at considerably lower potential than its non-SMATed counterpart. However, such behaviour was absent in the AISI 2205 steel. XPS analysis confirmed that the stoichiometry of Cr-oxide formed on the surface of SMATed AISI 304L steel was different than its non-SMATed surface. At the surface of the SMATed AISI 2205 steel, Cr-oxide formed was thicker than its non-SMATed surface.

(4) The plasma-nitriding response of the SMATed and non-SMATed AISI 2205 steels was different than the AISI 304L steel. SMATed and non-SMATed specimens of AISI 2205 steel showed the negligible change in the thickness of the nitrided layer. However, the nitrided layer was thicker on the SMATed surface of AISI 304L steel than its non-SMATed surface. This behaviour of the nitriding kinetics was linked with the nature of passive film.

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